

Dimitri Chrysafis

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Education

University of Wisconsin–Madison

August 2025 - May 2027

Bachelor of Science in Computer Science

Relevant Coursework: COMP SCI 577 (Algorithms), COMP SCI 566 (Computer Vision), COMP SCI 574 (Data Management), ISYE 524 (Optimization), MATH 340 (Linear Algebra)

Technical Skills

Languages: Python, C++, Swift, Julia, Go, JavaScript

Tools: MLX, PyTorch, WebGPU/WGSL, WebGL/GLSL, Metal, SwiftUI, JuMP/HiGHS

Projects

Qwen3.8-Flash-Next NVMe Inference Engine [\[GitHub\]](#)

Aug–Sep 2026

- Built a disk-backed inference engine that runs the **180B-parameter Qwen3.8-Flash-Next** model on a Mac with **36 GB unified memory** by streaming MoE experts from NVMe instead of loading the full model into memory
- Implemented memory-mapped weights, adaptive expert caching, and asynchronous prefetching; measured **11.3 GiB peak MLX allocation** during a 64-token run on M3 Max with an 8 GiB expert cache

Quantum Circuit Routing [\[Final Report\]](#)

Jul 2026

- Built an exact Julia/JuMP/HiGHS MILP that routes logical circuits onto sparse superconducting hardware via SWAPs, with calibrated coupling and readout errors
- Solved all 24 experiment instances to proven optimality; a 3×3 grid topology cut routing cost **56.1%** versus the 7-qubit heavy-hex baseline

Interactive 3D Fluid Simulator (MLS-MPM) [\[Demo\]](#) [\[GitHub\]](#)

Sep 2025–Aug 2026

- Engineered a browser-based 3D MLS-MPM simulator in JavaScript/WebGPU, simulating **400,000 particles** on an M3 Max with GPU-resident state and staged compute kernels
- Implemented fixed-point atomic accumulation for particle-to-grid transfers plus interactive dam-break, piston, particle-injection, and orbit-camera controls

Zwift Shifter (macOS Virtual Gear Shifter) [\[GitHub\]](#) [\[Blog\]](#)

Aug 2026

- Reverse-engineered Zwift’s Bluetooth shifter with LLDB and injected a dylib to feed synthetic events into its native gear-shifting logic
- Shipped a SwiftUI app that changes live Zwift HUD gears over a local Unix socket, verified on an Apple Silicon Mac

GPU Fractal Rendering Engine [\[GitHub\]](#) [\[Blog\]](#)

Feb 2025

- Developed GPU fractal renderers in WebGL2/GLSL and Apple Metal for Newton, Kleinian, and Julia sets with interactive pan/zoom and shader-based iteration
- Implemented a Metal compute pipeline for **6,000×6,000** Julia frames with adaptive iteration counts and GPU-parallel image generation

Additional

Ironman Wisconsin finisher: 2.4 miles swimming, 112 miles cycling, 26.2 miles running